

PFC Boost Inductor — Production Engine Equation Reference (v11)

Step-by-step equations for every quantity computed by our production backend `step7_magnetic_calc.py` (Phase-1 v10 + Phase-B improvements). Transcribed from the code as of 2026-06-13. **Updates since the v10 reference (2026-06-10):** material-specific DB DC-bias $k(H)$ in the per- θ waveform loop (no EDGE fallback); harmonic AC-excess copper loss (K_{harm}); discontinuous-conduction (DCM) detection; instantaneous-peak minimum-L guarantee; and the single-source view contract feeding Result / Review / Simulation. Counterpart record to the Simulation-Agent's `PFC_Inductor_Engine_Equations.pdf`.

Conventions. Distributed-gap powder toroid (ferrite path noted where it differs), CCM boost PFC, n_{ph} -phase interleaved. Geometry in mm; flux density in T; bias field H in A/m internally and oersted (Oe) for retention; Steinmetz frequency in Hz into the DB; resistance in Ω ; power in W; temperature in $^{\circ}\text{C}$. $\theta \in (0, \pi)$ is the line half-cycle angle; the engine integrates θ -dependent quantities on an **M = 360-point** midpoint grid. Subscript “single” = one core of the stack; stack totals = single $\times n_{\text{stk}}$. **Material data (core loss $P_v(B, f, T)$ and DC-bias $k(H)$) comes from the MagnetisDB as temperature-aware interpolated curves**, not closed-form fits — this is our key accuracy lever and is shown below as $\text{DB}(\cdot)$.

S0 — Physical & model constants

(E1) $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$, $\rho_{\text{Cu},20} = 1.72 \times 10^{-8} \Omega \cdot \text{m}$, $\alpha_{\text{Cu}} = 0.00393 \text{ K}^{-1}$

- copper resistivity at 20 $^{\circ}\text{C}$ and its temperature coefficient

- $\rho_{\text{Cu}}(T) = \rho_{\text{Cu},20}(1 + \alpha_{\text{Cu}}(T - 20))$

(E2) $1 \text{ Oe} = 79.577 \text{ A/m}$

- oersted conversion used wherever the DB bias curve is addressed in Oe

(E3) $i_{\text{GSE}}: c = 1.444, b = 2.106, I_c = 3.5435, K_{i_{\text{GSE}}} = 2^c / ((2\pi)^{c-1} \cdot I_c)$

- fixed improved-Generalised-Steinmetz constants for the triangular-ripple duty correction $F(D)$

(E4) $K_{\text{harm}} = 1.213$ (triangular-ripple harmonic AC-excess factor)

- amplifies ONLY the AC excess ($R_{\text{AC}}/R_{\text{DC}} - 1$) of the HF copper loss: the switching ripple's higher harmonics ($n = 1, 3, 5, \dots$) see a higher effective R_{AC}

Note: NEW (v11): adopted from the simulation-agent engine. $K_{\text{harm}} = 1$ recovers the v10 form $I_{\text{hf,rms}}^2 \cdot R_{\text{AC}}$.

(E5) Proximity (Dowell-like, v10): $k_{\text{skin}} = 0.50, k_{\text{prox}} = 0.40, k_{\text{crowd}} = 0.25$

- calibrated coefficients for the litz/TIW AC-resistance model (E20)

(E6) Thermal 2-node split (v10): $s_c = 1.00, s_w = 0.90, \text{couple} = 0.50, \text{hotspot} = 1.12; I_{\text{lead}} = 150 \text{ mm}$

- core/winding ambient-path split, core-winding coupling, interior hotspot factor; default lead length added to copper length

S1 — Unbiased inductance, required retention, and turns (powder)

(E7) $L_0 = A_{L,\text{single}} \cdot n_{\text{stk}} \cdot N^2 \times 10^{-9} [\text{H}]$

- $A_{L,\text{single}}$ — inductance factor of ONE core at zero bias [nH/T^2]; N — turns

(E8) $L_{0,\text{min}} = L_0(1 - \text{tol}_{\text{AL}}), L_{0,\text{max}} = L_0(1 + \text{tol}_{\text{AL}})$ ($\text{tol}_{\text{AL}} \approx 0.08$)

(E9) $k_{\text{req}} = L_{\text{target}} / L_0$ (required bias retention at the worst DC point)

- turns are chosen as the smallest N whose biased inductance still meets the target:

- find N : $\text{DB } k(H_{\text{worst}}(N)) \cdot L_0(N) \geq L_{\text{target}}$

(E10) $I_{\text{dc,worst}} = \max_{\text{OPS}} [\sqrt{2} \cdot P_{\text{out}} / (\eta \cdot V_{\text{in}} \cdot \text{PF} \cdot n_{\text{ph}})]$

- worst-case per-phase crest current scanned over all 9 operating points (OPS)

Note: Ferrite path instead converges turns + gap l_g to a target peak flux (E14 uses the gapped reluctance).

S2 — DC-bias field and DB permeability retention

(E11) $H_{\text{dc}} = N \cdot I / l_{e,\text{single}} [\text{A/m}], H_{\text{Oe}} = H_{\text{dc}} / 79.577$

- uses l_e of ONE core (NOT $n_{\text{stk}} \cdot l_e$) for stacked toroids

(E12) $k(H) = \text{DB.get_k_bias}(\text{material}, H_{\text{Oe}})$ [–]

- material-specific DC-bias curve interpolated from the MagneticsDB (per material, per H)

Note: FIXED (v11): the per- θ waveform loop now ALSO calls $\text{DB.get_k_bias}(\text{material}, H_{\text{Oe}})$ for the SELECTED material — the v10 EDGE-calibrated closed-form fallback $k(H) = \min(1, [1/(0.01 + 9.202 \times 10^{-10} H^{3.044})]/100)$ is no longer used there. This corrects $k(H)/B_{\text{max}}$ for non-EDGE materials and makes step7's biased flux match the field-engine, which reads the same DB curve.

(E13) $L(H) = k(H) \cdot L_0$ (biased inductance)

S3 — Operating-point currents (per OPS corner $\{V_{\text{in}}, P_{\text{out}}, \eta, \text{PF}\}$)

(E14) $P_{\text{in}} = P_{\text{out}} / \eta$, $V_{\text{pk}} = \sqrt{2} \cdot V_{\text{in}}$, $D_{\text{pk}} = 1 - V_{\text{pk}} / V_{\text{out}}$

- $V_{\text{out}} = 393$ V boost rail; D_{pk} — duty at the line crest

(E15) $I_{\phi, \text{avg}, \text{crest}} = \sqrt{2} \cdot P_{\text{out}} / (\eta \cdot V_{\text{in}} \cdot \text{PF} \cdot n_{\text{ph}})$

- per-phase switching-average current at the line crest

S4 — Line-cycle waveforms over $\theta \in (0, \pi)$, $M = 360$ midpoints

(E16) $\theta_n = (n + 1/2)\pi/M$, $v_{\text{in}}(\theta) = V_{\text{pk}} \sin\theta$, $D(\theta) = \text{clip}(1 - v_{\text{in}}/V_{\text{out}}, 0, 0.98)$

(E17) $i_{\text{avg}}(\theta) = I_{\phi, \text{crest}} \sin\theta$, $H(\theta) = 0.4\pi \cdot N \cdot i_{\text{avg}}(\theta) / I_{\text{e}, \text{cm}}$, $L(\theta) = k(H(\theta)) \cdot L_0$

- powder: $k(H(\theta)) = \text{DB.get_k_bias}(\text{material}, H(\theta))$ per θ (v11 — was an EDGE closed-form in v10)

(E18) $\Delta I_{\text{pp}}(\theta) = v_{\text{in}}(\theta) \cdot D(\theta) / (L(\theta) \cdot f_{\text{sw}})$, $I_{\text{hf}}(\theta) = \Delta I_{\text{pp}}(\theta) / (2\sqrt{3})$

- triangular switching ripple peak-to-peak and its RMS

(E19) DCM at θ if $i_{\text{avg}}(\theta) < \Delta I_{\text{pp}}(\theta)/2$; $f_{\text{DCM}} = (\text{count of such } \theta) / M$

- NEW (v11): fraction of the line half-cycle where the inductor current rings to zero (boundary / discontinuous conduction) — reported as dcm_fraction

(E20) $B_{\text{ac}, \text{pk}}(\theta) = v_{\text{in}}(\theta) \cdot D(\theta) / (2 \cdot N \cdot A_{\text{e}} \cdot f_{\text{sw}})$, $B_{\text{dc}}(\theta) = L(\theta) \cdot i_{\text{avg}}(\theta) / (N \cdot A_{\text{e}})$, $B_{\text{max}}(\theta) = B_{\text{dc}} + B_{\text{ac}, \text{pk}}$

- $A_{\text{e}} = A_{\text{e}, \text{single}} \cdot n_{\text{stk}} \times 10^{-6} [\text{m}^2]$

S5 — Flux density, radial crowding, saturation margin

(E21) $B_{\text{ac}, \text{pk}} = V_{\text{pk}, 90} \cdot D_{\text{pk}, 90} / (2N \cdot A_{\text{e}} \cdot f_{\text{sw}})$, $B_{\text{dc}} = L_{\text{target}} \cdot I_{\phi, \text{crest}} / (N \cdot A_{\text{e}})$

- headline flux evaluated at the 90 Vac low-line reference corner

(E22) $B_{\text{max}, \text{FL}} = B_{\text{dc}} + B_{\text{ac}, \text{pk}}$, $B_{\text{min}, \text{FL}} = B_{\text{dc}} - B_{\text{ac}, \text{pk}}$

(E23) $\text{crowd}_{\text{ax}} = r_{\text{mean}} / r_{\text{in}} = [(ID/2 + OD/2)/2] / (ID/2)$, $B_{\text{inner}} = B_{\text{max}, \text{FL}} \cdot \text{crowd}_{\text{ax}}$

- toroid $1/r$ inner-bore flux concentration (v10)

(E24) $\text{margin}_{\text{sat}} = (B_{\text{sat}}(T_{\text{core}}) - B_{\text{max}}) / B_{\text{sat}} \times 100 \%$

- $B_{\text{sat}}(T_{\text{core}})$ from the DB at the converged core temperature; inner-bore margin checked too

(E25) $L_{\text{full}, \text{min@pk}} = L_{0, \text{min}} \cdot \text{DB.k}(H_{\text{pk}})$, $H_{\text{pk}} = N \cdot (I_{\phi, \text{crest}} + \Delta I_{\text{pp}}/2) / (I_{\text{e}, \text{single}} \cdot 79.577) [\text{Oe}]$

- NEW (v11): minimum guaranteed inductance at the worst INSTANTANEOUS peak bias (crest-average current + half the ripple), via the selected material's DB curve at $\text{min-}A_{\text{L}}$

Note: Informational — more conservative than the crest-average bias; turns selection and pass/fail are unchanged. Powder only.

(E26) Ferrite: $B_{\text{dc}} = \mu_0 N \cdot I / (l_{\text{g}} + l_{\text{e}}/\mu_r)$

- gapped-reluctance bias flux; $\mu_r = \text{DB.get_mu_r}(\text{material}, 100^\circ\text{C})$; l_{g} — air gap

S6 — Copper geometry, MLT, and DC resistance

(E27) $w_{\text{core}} = (OD - ID)/2$, $\text{MLT} = 2(w_{\text{core}} + HT \cdot n_{\text{stk}}) + 2 \cdot OD_{\text{bundle}} [\text{mm}]$ (v10)

- v10 uses the actual bundle OD as the build allowance (legacy form used a fixed 3.8 mm)

(E28) $I_{\text{Cu}} = N \cdot \text{MLT}/1000 + I_{\text{lead}}/1000 [\text{m}]$

- lead wire (entry + exit, 150 mm) added to the wound length

$$(E29) R_{DC}(T) = R'_{20} \cdot (1 + \alpha_{Cu}(T - 20)) \cdot l_{Cu} \text{ [}\Omega\text{]}$$

- R'_{20} — catalog wire resistance per metre at 20 °C [Ω/m]

$$(E30) FF_{Cu} = N \cdot A_{Cu} / W_{a,bore}$$

- bare-copper bore-window fill; $W_{a,bore}$ is the single-core window area (invariant with stacks)

S7 — Skin depth and AC resistance (physics-based Dowell-proximity, v10)

$$(E31) \delta = \sqrt{(\rho_{Cu}(T) / (\pi \cdot f_{sw} \cdot \mu_0))}, x = d_{strand} / (2\delta)$$

- δ — skin depth at f_{sw} ; x — strand-radius / skin-depth ratio

$$(E32) F_{skin} = 1 + k_{skin} \cdot x^2, F_{prox} = 1 + k_{prox} \cdot (N_{lay} - 1) \cdot x^2 \cdot (1 + k_{crowd}(\text{crowd}_{ax} - 1))$$

- inter-layer proximity grows with layer count and inner-bore crowding

$$(E33) R_{AC}/R_{DC} = \max(1, F_{skin} \cdot F_{prox})$$

- litz/TIW path; solid/enamel uses an exact Bessel skin factor (compute_dowell_factor)

Note: This physics-based ratio is a strength of our engine vs. a flat 1.15 heuristic.

S8 — Bore layer count and residual window

$$(E34) OD_{bundle} \text{ — catalog bundle OD (authoritative) or computed from strands \& fill}$$

- feeds both MLT (E15) and the layer/proximity model (E22)

$$(E35) N_{lay}, t_{pl}, r_{bore} = \text{layers}(N \cdot n_{par}, ID, OD_{bundle}); \text{hole}_{ID} = \max(0.5, 2 \cdot r_{bore})$$

- turns-per-bore-layer and residual bore radius after all layers (negative □ overfull)

S9 — Core loss (DB Steinmetz, iGSE duty correction, cycle-averaged)

$$(E36) P_v(B, f, T) = \text{DB.get_core_loss}(\text{material}, f_{sw}, B_{ac,pk}, T) \text{ [W/m}^3\text{]}$$

- temperature-aware loss density interpolated from the MagneticsDB material curves

$$(E37) F_D(D) = K_{iGSE} \cdot (D^{1-c} + (1-D)^{1-c}), D \text{ clipped to [0.02, 0.98]}$$

- iGSE duty-ratio correction for the asymmetric triangular flux

$$(E38) p_{core}(\theta) = P_v(B_{ac,pk}(\theta), f_{sw}, T_{core}) \cdot F_D(D(\theta)) \cdot V_e, P_{core} = (1/M) \sum_n p_{core}(\theta_n)$$

- $V_e = V_{e,single} \cdot n_{stk}$ [m³]; this cycle-averaged value is the authoritative P_{core}

$$(E39) \text{Uncertainty band: } P_{unc,lo} = P_{Cu,100} + 1.05 \cdot P_{core}, P_{unc,hi} = P_{Cu,100} + 1.20 \cdot P_{core}$$

- +5 % to +20 % core-loss uplift band (unanchored until FEA/bench calibration)

S10 — RMS decomposition and copper loss (cycle-averaged)

$$(E40) I_{\phi,rms} = \sqrt{(1/M) \sum i_{avg}(\theta)^2}, I_{hf,rms} = \sqrt{(1/M) \sum i_{hf}(\theta)^2}$$

$$(E41) P_{Cu}(T) = I_{\phi,rms}^2 \cdot R_{DC}(T) + I_{hf,rms}^2 \cdot [R_{DC}(T) + (R_{AC}(T) - R_{DC}(T)) \cdot K_{harm}]$$

- DC-resistance part on the full RMS, plus the AC EXCESS ($R_{AC} - R_{DC}$) on the ripple RMS, amplified by the harmonic factor $K_{harm} = 1.213$

- $= I_{\phi,rms}^2 R_{DC} + I_{hf,rms}^2 R_{DC} (1 + (R_{AC}/R_{DC} - 1) K_{harm})$; reported at T_{core} , 25 °C and 100 °C

Note: CHANGED (v11): v10 used $I_{hf,rms}^2 \cdot R_{AC}$. The harmonic uplift raises HF copper loss to reflect the triangular ripple's spectrum.

$$(E42) J = I_{\phi,rms} / (A_{Cu} / n_{par}) \text{ [A/mm}^2\text{]}$$

- copper current density per parallel conductor

S11 — Thermal (SA single-node convergence + v10 two-node hotspot)

$$(E43) SA = [\pi \cdot OD_w \cdot OH + (\pi/2)(OD_w^2 - \text{hole}^2) + \pi \cdot \text{hole} \cdot OH] / 100 \text{ [cm}^2\text{]}$$

- wound-part exposed surface area; OD_w , OH from the assembled wound envelope

$$(E44) \Delta T_{SA} = (P_{tot} \cdot 1000 / SA)^{0.833} \text{ [K]}$$

- natural-convection surface-area power law

(E45) **Convergence:** iterate $T_{\text{core}} \leftarrow T_{\text{amb}} + \Delta T_{\text{SA}}(P_{\text{core}}(T_{\text{core}}) + P_{\text{Cu}}(T_{\text{core}}) + P_{\text{fring}})$ until $|\Delta| < 0.2 \text{ K}$

- couples temperature-dependent core & copper loss to the surface model (10-iteration loop)

(E46) $\theta = \Delta T_{\text{SA}} / P_{\text{tot}}$; $R_{\text{ca}} = \theta(s_{\text{c}} + s_{\text{w}}) / s_{\text{c}}$, $R_{\text{wa}} = \theta(s_{\text{c}} + s_{\text{w}}) / s_{\text{w}}$, $R_{\text{cw}} = \theta \cdot \text{couple}$

- two-node resistances derived from the calibrated SA baseline

(E47) Solve KCL 2x2 for ΔT_{c} , ΔT_{w} ; $\Delta T_{\text{hotspot}} = \max(\Delta T_{\text{c}}, \Delta T_{\text{w}}) \cdot 1.12$

- core/winding node rises and interior hotspot (1.12 over node average)

S12 — Gap fringing (ferrite only)

(E48) $P_{\text{fring}} = \text{Rogowski}(I_{\text{g}}, A_{\text{e}}, W_{\text{a}}, N, d_{\text{bundle}}, R_{\text{DC},100})$

- gap-fringing eddy loss in winding turns near the gap (compute_rogowski_fringing); powder cores: 0

S13 — Per- V_{in} screening table (analytical, single-point)

A faster single-point analytical pass over all 9 OPS corners drives the per- V_{in} sweep table and charts. It is intentionally lighter than the S4/S9/S10 waveform integration:

(E49) $B_{\text{ac,pk}} = V_{\text{pk}} D_{\text{pk}} / (2N \cdot A_{\text{e}} f_{\text{sw}})$, $P_{\text{core}} = P_{\text{v}}(B_{\text{ac,pk}}, f, T) \cdot F_{\text{D}}(D_{\text{pk}}) \cdot V_{\text{e}}$, $P_{\text{Cu}} = I_{\text{rms}}^2 R_{\text{DC}}(T) + I_{\text{hf}}^2 R_{\text{AC}}(T)$

- single crest-point evaluation per corner (no θ integration)

Note: IMPORTANT: this table is a SCREENING estimate and differs from the authoritative S4/S9/S10 waveform totals at the same V_{in} . The Review page anchors this table to the waveform result at 90 V / 100 °C so the two presentations agree (P_{core} , P_{tot} ratios).

S14 — Worst-case scan, acceptance, and ranking score

Every OPS corner is evaluated; the design reports worst-case total loss, B_{max} (mean path) and B_{inner} , minimum guaranteed L, maximum ΔT and J. Pass/fail uses $L \geq L_{\text{target}}$, $B_{\text{max}} \leq B_{\text{sat}}(T)$, $FF_{\text{Cu}} \leq \text{limit}$, $\Delta T \leq \text{budget}$, J in band.

(E50) $\text{score} = w_{\text{loss}} \cdot P_{\text{tot}} / P_{\text{ref}} + w_{\text{vol}} \cdot V_{\text{e}} / V_{\text{ref}} + w_{\Delta T} \cdot \Delta T / \Delta T_{\text{ref}} + w_{\text{cost}} \cdot (...) + w_{\text{fill}} \cdot (...)$

- composite ranking score (lower is better); candidates returned top-N per stack count

S15 — Single-source view contract (Result / Review / Simulation)

NEW (v11). build_view_contract(result, state) is the AUTHORITATIVE render payload every screen draws from. It re-runs the S4/S9/S10 waveform integration on the STORED design so all three screens render the exact same converged numbers — they RENDER this, they never recompute. Additive / read-only.

(E51) **waveform = _half_cycle_averages(..., $V_{\text{in,pk}} = \sqrt{2} \cdot 90$, $R_{\text{dc}} = R_{\text{DC},100}$, $M = 360$, series)**

- the 90 Vac DESIGN-corner per- θ arrays (t , v_{in} , D , i_{avg} , H , B_{dc} , $B_{\text{ac,pk}}$, B_{max} , I_{hf} , P_{core} , P_{Cu} , P_{tot}) on the 100 °C copper basis (so Σp_{cu} equals reported $P_{\text{Cu},100}$)

(E52) $i_{\text{crest}}(V_{\text{in}}) = I_{\phi, \text{crest}} \cdot (I_{\text{crest}}^{\text{sweep}}(V_{\text{in}}) / I_{\text{crest}}^{\text{sweep}}(90))$, $M = 180$

- per-OPS waveforms for the V_{in} explorer, anchored so the design corner equals the readouts

(E53) $P_{\text{core}}^{\text{anc}}(V_{\text{in}}) = P_{\text{core}}^{\text{tbl}}(V_{\text{in}}) \cdot (P_{\text{core}}^{\text{design}} / P_{\text{core}}^{\text{tbl}}(90))$, $P_{\text{Cu}}^{\text{anc}}$ likewise

- the single-point S13 screening table is rescaled to the authoritative S4/S9/S10 design totals at 90 V, so the per- V_{in} sweep agrees with Result / Review at every corner

(E54) **acceptance = { $B_{\text{max}} < B_{\text{sat}}(T)$, $K_{\text{u}} \leq 60 \%$, $\Delta T \leq \text{budget}$, $L_{\text{full}} \geq L_{\text{target}}$ } $\square$ APPROVE / REJECT**

- step7's own verdict + fail reasons exported for the viewer's acceptance panel; limits come from the design, never invented

Appendix — Variable glossary (code names)

Symbol	Code name (step7_magnetic_calc.py)	Unit	Meaning
μ_0	MU0	H/m	Permeability of free space
ρ_{Cu}, α_{Cu}	RHO_CU_20 / ALPHA_CU	ohm*m, 1/K	Copper resistivity / temp coefficient
c,b,lc,K	IGSE_C / IGSE_B / IGSE_IC / IGSE_K	-	iGSE duty-correction constants
Kharm	K_HARM	-	Triangular-ripple harmonic AC-excess factor (1.213, v11)
kskin,kprox,kcrowd	_PROX_kSkin/kProx/kCrowd	-	Dowell-proximity coefficients (v10)
sC,sW,couple,hotspot	_THERM_sC/sW/couple/hotspot	-	Two-node thermal split + hotspot factor
N	N	turns	Number of turns (chosen by _turns_powder/_turns_ferrite)
nstk	stacks	-	Stacked core count
AL,single	AL_nom_nH	nH/T2	Inductance factor of ONE core
L0(min/nom/max)	L0_min/nom/max_uH	uH	Unbiased inductance + AL tolerance band
kreq	kreq_nom	-	Required retention L_target/L0
Ae,single/Ve,single	Ae_single_mm2 / Ve_total_cm3	mm2/cm3	Single-core area / total volume
Le,single	Le_single_mm	mm	Magnetic path length of ONE core
Wa,bore	Wa_single_mm2	mm2	Single-core bore window area
Bsat(T)	Bsat_at_Tcore	T	Saturation flux at core temperature (DB)
k(H)	DB.get_k_bias / _retention_edge	-	DC-bias retention (DB curve; EDGE fallback)
Pv(B,f,T)	DB.get_core_loss	W/m3	Temperature-aware core-loss density (DB)
Hdc/HOe	H_Oe_design / H_Oe_worst	A/m, Oe	DC bias field
Iphi,crest	I_phi_avg_crest	A	Per-phase crest average current
Dpk	Dpk90	-	Duty at line crest
dLpp/lhf	dL_pp_A / lhfRms_A	A	Switching ripple pk-pk / RMS
Bac,pk/Bdc/Bmax	Bac_pk_T / Bdc_T / Bmax_FL_T	T	AC peak / DC / mean-path peak flux
crowd_ax/Binner	crowd_axial / Bmax_inner_FL_T	-, T	Inner-bore crowding / crowded peak flux
MLT	MLT_v10_mm	mm	Mean length per turn (2(w+HT*stk)+2*ODb)
ICu	Cu_length_m	m	Total conductor length incl. leads
RDC(T)	DCR_25C/100C_mOhm	mohm	DC resistance at temperature
RAC/RDC	Rac_Rdc	-	AC/DC ratio (Dowell-proximity or Bessel)
delta/x	(internal)	m, -	Skin depth / strand-to-skin ratio
Nlay,tpl,rbore	layers_needed / (internal)	-	Bore layers, turns/layer, residual radius
FFcu/Ku	FFcu / Ku	-	Window fill / bore utilization
FD(D)	_Fd(D)	-	iGSE duty correction
Pcore	Pcore_W	W	Cycle-averaged core loss (authoritative)
Punc lo/hi	P_unc_lo_W / P_unc_hi_W	W	Core-loss uncertainty band
fDCM	dcm_fraction	-	Fraction of half-cycle in DCM, i_avg<dLpp/2 (v11)
Lfull,min@pk	Lfull_min_at_peak_uH	uH	Min-L at worst instantaneous peak bias, DB k(H) (v11)
Iphi,rms/Pcu	Ihf_rms_A / Pcu_100C_W	A, W	RMS current / copper loss
J	J_A_mm2	A/mm2	Copper current density
SA/dT	SA (internal) / dT_rise_C	cm2, K	Wound surface area / surface rise
dTc/dTw/dThot	dT_core_C/dT_wdg_C/dT_hotspot_C	K	Two-node rises + hotspot
Rca/Rwa/Rcw	(returned by _two_node_thermal)	K/W	Node thermal resistances
Pfring	P_fringing_W	W	Gap fringing loss (ferrite)
Ptotal(25/100)	Ptotal_25C_W / Ptotal_100C_W	W	Total loss at temperature

Symbol	Code name (step7_magnetic_calc.py)	Unit	Meaning
score	score	-	Composite ranking score (lower better)

Honest status note (v11). Strengths vs. the Simulation-Agent reference engine: temperature-aware DB core-loss and DC-bias curves, now used per- θ in the waveform loop for the **SELECTED** material (the v10 EDGE-calibrated $k(H)$ fallback is removed); a physics-based Dowell-proximity R_{AC} with the $K_{harm} = 1.213$ triangular-ripple harmonic uplift on the AC excess; a two-node hotspot thermal model; ferrite gap fringing; DCM-fraction detection; an instantaneous-peak minimum-L guarantee; and a single-source view contract (S15) so Result / Review / Simulation render identical numbers. Open items: the per- V_{in} screening table (S13) is single-point and is anchored to the waveform totals at 90 V; R_{AC} (incl. K_{harm}), thermal and bias retention are still first-order until FEA/bench data is attached. All other quantities are exact arithmetic on the DB + designer-selected geometry/wire.